MODULE 22 · DOMAIN 3 - IMPLEMENT COMPUTER VISION SOLUTIONS

Generating and editing video

Sora 2 video jobs, sizes and durations, reference images, remix, operational limits, content restrictions, and an honest note on lifecycle.

DOMAIN 3 - Computer vision WEIGHT 10-15% MODULE 22 of 39

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Sora 2 is a preview with a short life

THE DATES

Retiring soon

- Sora 2 on Azure is in preview
- Version 2025-12-08 retires 15 October 2026
- No replacement is listed yet
- OpenAI's own Videos API shuts down 24 September 2026

WHAT LASTS

The workflow

- Video is an asynchronous job
- Reference inputs anchor the result
- Remix edits instead of regenerating
- Limits and content rules shape designs

ON THE EXAM

Check the model retirement schedule before building. Learn the patterns here; they outlast any one model.

A video is a **job**, not a response. You **start** it, **check** on it, then **collect** it, before it expires.

What Sora 2 can do

- **Text to video**, **image to video**, and **generated video to video** through remix.
- **Audio** is generated as part of the output video.
- Uses the Azure OpenAI **v1 API**, with the same structure as OpenAI's video API.
- The **Video playground** in the Foundry portal shows matching Python and cURL code.
- Billing is **per second** of generated video.

> ON THE EXAM
>
> If the SDK says the client has no 'videos' attribute, upgrade the openai package.

Five endpoints

ENDPOINT	WHAT IT DOES
`Create video`	Starts a render job from a prompt, a reference image, or a remix ID
`Get video status`	Returns the job's current state and progress
`Download video`	Fetches the finished MP4 once the job completes
`List videos`	Pages through your videos, for history or housekeeping
`Delete video`	Removes a video from Azure OpenAI's storage

ON THE EXAM

Create returns an ID straight away, not a video. Everything after that uses the ID.

From prompt to MP4

01
Create
Send prompt and settings; get an ID.

02
Queued
Waiting for capacity.

03
In progress
Rendering, often 1 to 5 minutes.

04
Completed
Or failed, with a reason.

05
Download
Fetch the MP4 content.

06
Store
Keep your own copy.

ON THE EXAM

Poll with a pause between checks. Microsoft's sample waits 20 seconds each time.

Create, poll, download

```
video = client.videos.create(
    model="sora-2",
    prompt="Slow dolly shot of a ceramic mug on a sunlit kitchen table",
    size="1280x720",
    seconds="8")

while video.status not in ("completed", "failed"):
    time.sleep(20)
    video = client.videos.retrieve(video.id)

if video.status == "completed":
    client.videos.download_content(video.id, variant="video").write_to_file("mug.mp4")
```

ON THE EXAM

Downloading before the status is completed fails. The loop is the lab window you keep checking.

Sora 2 parameters

PARAMETER	VALUES	NOTES
`size`	720x1280 or 1280x720	Portrait is the default
`seconds`	4, 8 or 12	Default 4; sent as a string
`input_reference`	One JPEG, PNG or WebP image	Anchors the first frame; must match size
`remix`	A completed video's ID	Edits instead of regenerating
`prompt`	Shot, subject, action, setting, light, camera	Keep it single-purpose

ON THE EXAM

Durations are fixed choices, not any number. Longer pieces are built from several clips.

Shorter clips follow instructions **more reliably**. Two 4-second clips, stitched, often beat one 8-second clip.

Writing a video prompt

- Describe the **shot type**, **subject**, **action**, **setting**, **lighting** and **camera motion**.
- Keep each prompt **single-purpose**: one scene, one main action.
- Write in **English or another Latin-script language** for the best results.
- The model can struggle with **complex physics** and **cause and effect**, like bite marks on a cookie.
- It can also confuse **left and right**, and precise **timing** of events or camera moves.

> ON THE EXAM
>
> Known weaknesses are design constraints. Don't build an app that depends on precise left-right choreography.

Reference image or remix?

INPUT_REFERENCE

Start from a still

- One image anchors the first frame
- Creates a new video from it
- Image must match the video size exactly

REMIX

Edit a finished video

- Uses a completed video's ID
- Keeps structure, motion and framing
- Changes what the new prompt describes

THE TIEBREAKER

Bring a product photo to life: reference image. Change one thing in an approved clip: remix.

Remix done well

- Reference a **successfully completed** video's ID, and supply an **updated prompt**.
- Remix keeps the original's **framework, scene transitions and visual layout**.
- Limit each remix to **one clearly articulated adjustment**.
- Narrow, precise edits keep **more fidelity** and produce **fewer visual defects**.
- Several changes at once behave more like regenerating, and can lose what was approved.

> ON THE EXAM
>
> 'Keep everything, change the colour palette' is the ideal remix: one precise change, everything else preserved.

A reference image, then a remix

```
launch = client.videos.create(
    model="sora-2",
    prompt="The trainer slowly rotates as studio lights sweep across it",
    size="1280x720",
    seconds="4",
    input_reference=open("trainer_1280x720.png", "rb"))

# ...poll until launch.status is "completed"...

warmer = client.videos.remix(
    video_id=launch.id,
    prompt="Shift the colour palette to warm amber, keep everything else")
```

ON THE EXAM

The reference image is already 1280x720 to match size. The remix changes one thing only.

Limits that shape the design

- Documented limit: **two video jobs at a time**. A third must wait for one to finish.
- Jobs are available for **24 hours**. After that, the video must be generated again.
- So **download and store** finished videos promptly, in your own storage.
- Queue requests in your application rather than firing them all at once.
- **403**: assign Cognitive Services User. **Failed** job: read failure_reason.

> ON THE EXAM
>
> A batch of twenty videos needs a queue, and a pipeline that saves each MP4 as soon as it completes.

What Sora 2 won't generate

- **Real people**, including **public figures**, can't be generated.
- **Input images containing human faces** are currently rejected.
- **Copyrighted characters** and **copyrighted music** are rejected.
- Output is limited to content suitable for audiences **under 18**.
- Prompts, reference images and dialogue must all follow these rules, or the job fails.

> THE TIEBREAKER
>
> A requirement involving a real, identifiable person can't be met by any parameter. Redesign the concept.

Requirement to control

REQUIREMENT	CONTROL
A landscape web advert	size 1280x720
A vertical social clip	size 720x1280, the default
Start from a product photo	input_reference, resized to match exactly
Change one thing in an approved clip	remix with the video's ID
A longer sequence that stays on-brief	Several short clips, stitched
Many videos from one request	A queue: two jobs at a time
Keep the final video	Download and store within 24 hours

ON THE EXAM

Every row is a documented control or limit. Wrong answers usually invent a parameter that doesn't exist.

> A developer calls client.videos.create() for a Sora 2 deployment, then on the next line calls download_content() with the returned ID.
>
> The download fails every time, although the ID is valid and the job eventually succeeds.

What should the developer change?

A Increase the create call's timeout to five minutes so it returns the video.

B Poll the job's status until it is completed, and only then download.

C Request a higher quota so the video renders before the download call.

D Switch to the Images API, which returns the result in the same call.

◆ PAUSE AND DECIDE

What should the developer change?

A Longer create timeout.

Create returns a job ID immediately. It never returns the video, however long it waits.

B Poll, then download.

Video is an asynchronous job: download only works once status is completed.

C Higher quota.

Quota affects how many jobs run, not how long one takes to render.

D Images API.

Generates images, not video.

THE SIGNAL

'Valid ID, eventually succeeds, but download fails' means downloading before completion.

> A retailer animates product photos with Sora 2. Each request sets size to 1280x720 and sends the product photo, which is 1024x1024, as input_reference.
>
> Every request with a reference image is rejected.

What should the team change?

A Set seconds to 12, so there's more time to fit the square image in.

B Use remix with the product photo instead of a reference image.

C Resize or crop each photo to exactly 1280x720 before sending it.

D Convert each product photo to an animated GIF before uploading.

◆ PAUSE AND DECIDE

What should the team change?

A. Longer duration.
Duration has nothing to do with the reference image's dimensions.

B. Remix with the photo.
Remix takes a completed video's ID, not an image.

C. Match the image to 1280x720.
The reference image must match the video size exactly.

D. Animated GIF.
Reference images must be JPEG, PNG or WebP.

THE SIGNAL

A reference image rejected alongside a size setting: check that the two dimensions match exactly.

> Marketing approved an 8-second Sora 2 advert: its framing, motion and scene transitions are exactly right.
>
> They now want only a warmer colour palette. Regenerating from a new prompt keeps changing the framing and motion.

What should the team do?

A Create a new video with a longer, more detailed version of the prompt.

B Create a new video using a still of the advert's first frame as reference.

C Create four variants of the original prompt, then choose the warmest.

D Remix the approved video by its ID, with one colour palette change.

◆ PAUSE AND DECIDE

What should the team do?

A. Longer prompt, new video.
Still regenerates from scratch, so framing and motion can still change.

B. First frame as reference.
Anchors only the opening frame. Motion and transitions are generated again.

C. Four variants.
Four fresh generations, none of them the approved video.

D. Remix with one change.
Keeps structure, motion and framing, and changes only the palette.

THE SIGNAL

'Keep what was approved, change one thing' is the definition of a remix.

> A sports agency wants a Sora 2 clip of a famous footballer celebrating a goal. They plan to use a press photo of his face as the reference image.
>
> They ask which settings will make the result realistic.

What should you advise?

A Impossible: real people and faces are rejected. Change the concept.

B Leave out the photo, and describe the footballer by name in the prompt.

C Use the photo, and set seconds to 4 so the face appears only briefly.

D Use the photo, and remix the result to make the face more realistic.

◆ PAUSE AND DECIDE

What should you advise?

A Not possible; change the concept.

Real people, including public figures, can't be generated, and face images are rejected.

B Name him in the prompt.

Still generating a real public figure, which is blocked.

C A shorter clip.

Duration doesn't change the rule. The face image is rejected outright.

D Remix for realism.

The first job never succeeds, so there's nothing to remix.

THE TIEBREAKER

Content rules can't be tuned with parameters. A real, identifiable person means redesigning the idea.

Video in five lines

- **Video is a job:** create, poll until completed, download, store within 24 hours.
- **Sizes 720x1280 or 1280x720; 4, 8 or 12 seconds.** Short clips follow instructions best.
- **Reference image** anchors the first frame and must match size. **Remix** edits one thing.
- **No real people, faces, or copyrighted characters and music.** Two jobs at a time.
- **Sora 2 is a preview retiring in October 2026.** Check the schedule before building.

> ON THE EXAM
>
> Next, module 23: multimodal understanding, from captions and visual Q&A to Content Understanding.